

Statistics: How to Read a Timetable

Part of statistics in year 5 is learning about timetables. You'll learn how to read one and then how to interpret the information to answer questions based on the data it contains.

What are timetables?

Timetables are charts which show the departure and arrival times of transport, such as buses, trains and planes. They're very useful because a lot of information can be shown in a table. You will be able to work out lots of information, such as finding out when a plane is going to arrive, how long a train will take to get to a destination and which bus you need to get to arrive at your destination at a particular time.

Let's look at one:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

The table shows three buses that travel from Norwich to London. Each one has a different departure and arrival time.

Buses 11a and 11b both stop at all of the towns/cities but bus 11c doesn't stop at Newmarket.

This means that if you are trying to depart from Newmarket or arrive in Newmarket, you cannot get bus 11c.

We can answer questions about the times each bus stops at different places.

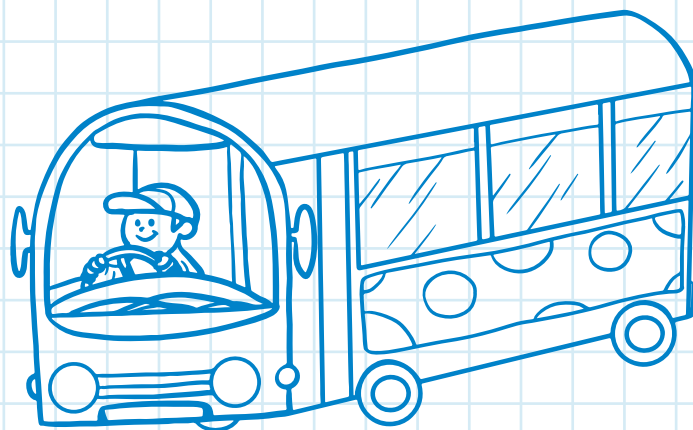
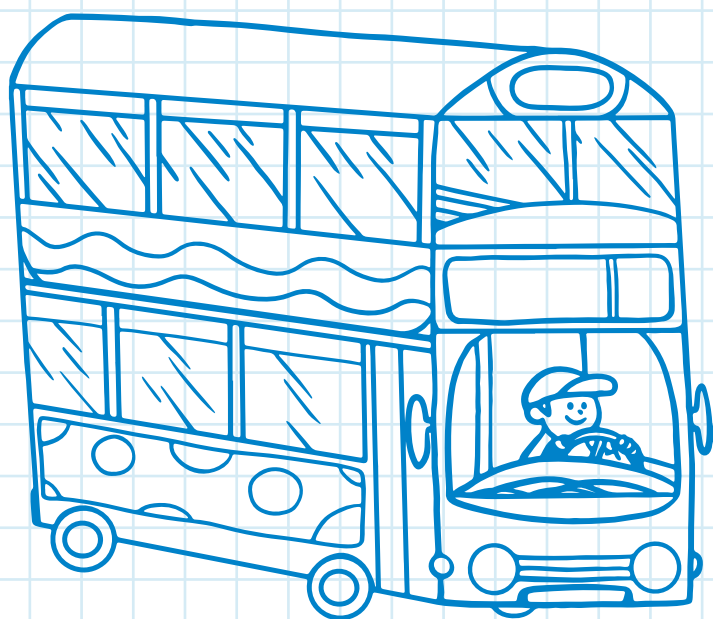
	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

At what time does bus 11b stop at Thetford?

To work this out, find the column showing the information for bus 11b, then find the destination of Thetford:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Bus 11b stops at Thetford at 08:30.



Which bus stops at Cambridge at 09:52?

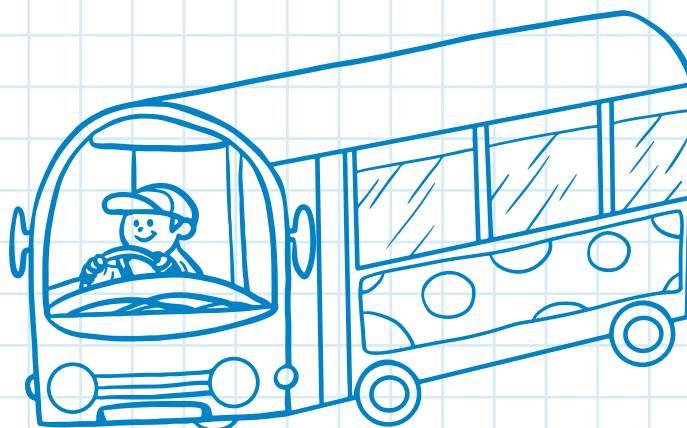
To work this out, find the information for Cambridge:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Now find the time 09:52 and see which bus column it is in:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Bus 11c stops at Cambridge at 09:52.



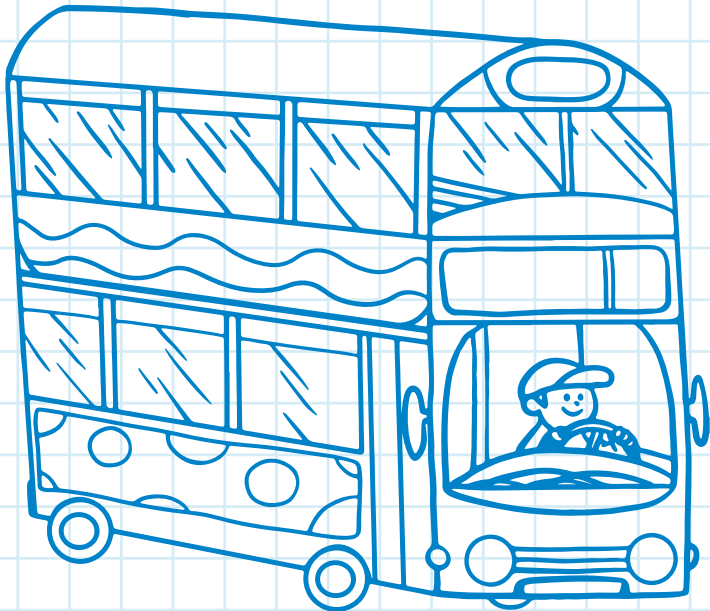
From the timetable, we can work out the duration (the length in time) of different journeys by finding the difference between the times.

How long does bus 11a take to get from Norwich to London?

Find what time the bus leaves Norwich and arrives in London:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Bus 11a departs Norwich at 07:10 and arrives in London at 10:32. This means the journey takes 3 hours 22 minutes.



How long does it take bus 11c to get from Norwich to Cambridge?

Find the time the bus leaves Norwich and arrives in Cambridge:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Bus 11c leaves Norwich at 08:15 and arrives in Cambridge at 09:52.

The journey takes 1 hour 37 minutes.

We can also compare times using the information from the timetables.

Which bus takes the longest to get from Cambridge to London?

To answer this, we need to look at the times each bus stops in Cambridge and the times they stop in London:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

We need to calculate the time each bus takes before we can work out the answer.

Bus 11a takes 1 hour 32 minutes.

Bus 11b takes 1 hour 31 minutes.

Bus 11c takes 1 hour 28 minutes.

From our calculations, we can see that bus 11a takes the longest to get from Cambridge to London.

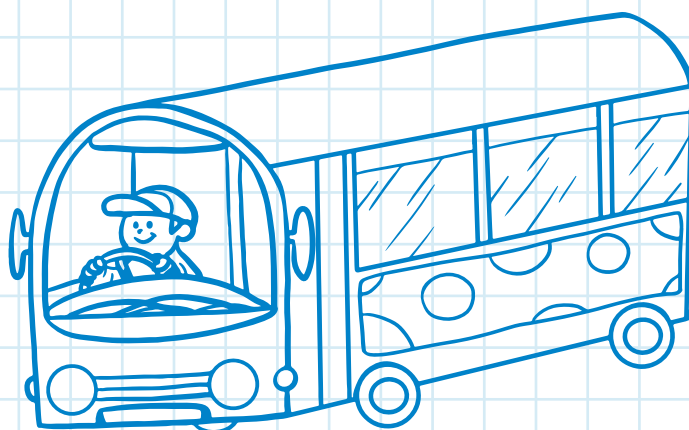
We can also work out problems which challenge us to find which bus someone might need to take.

**Cerys needs to catch a train to Brighton from London.
Her train leaves at 11:00.
Which bus should she get from Norwich to get there in time?**

To work this out, we need to look at the time each bus gets to London.
She has to be in London before 11.00 to give herself time to get the train:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Buses 11b and 11c will not get her in on time. Bus 11a gets in at 10:32, so she will have time to get her train at 11:00.



**Pritti lives in Thetford but works in Cambridge.
Her shift starts at 09:45.
What time should she get the bus and how long will the journey take?**

To work this out, we need to look at the times the buses arrive in Cambridge:

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

She can't get bus 11c as she won't be there in time.

She could get the bus that arrives in Cambridge at 09:00 or 09:34. Either one would be OK, but if she gets the one that arrives at 09:00, she has a long wait until she starts work. So she can get bus 11b. Now check the time the 11b stops in Thetford:

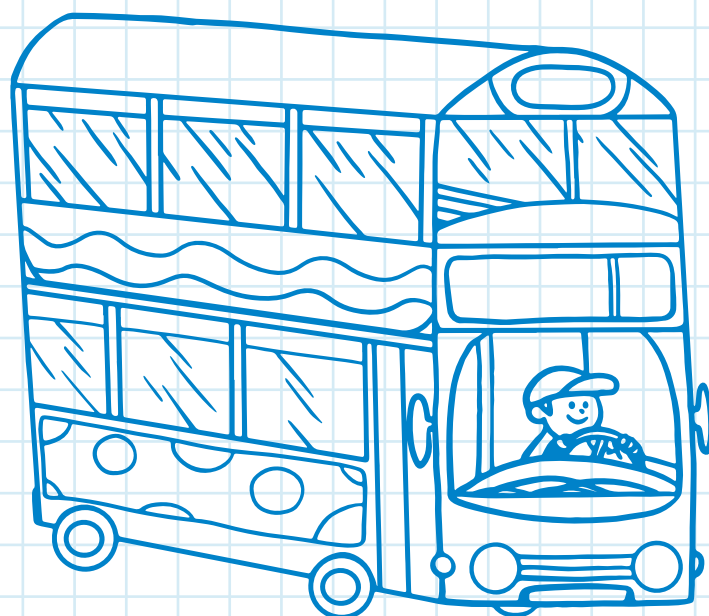
	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Pritti will need to get the bus at 08:30.

To work out how long the journey takes, find the time she leaves Thetford and the time she arrives at Cambridge (remember — she's travelling on bus 11b):

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

She gets on the bus at 08:30 and gets off at 09:34. The journey takes 1 hour and 4 minutes.



Bus Timetable Questions

	Bus 11a	Bus 11b	Bus 11c
Norwich	07:10	07:46	08:15
Thetford	07:55	08:30	09:00
Newmarket	08:29	09:02	-
Cambridge	09:00	09:34	09:52
London	10:32	11:05	11:20

Now use the same timetable to work out the questions.

1. Daryl takes the bus and arrives at Cambridge at 9 o'clock. Which bus does he take?

2. Hanya needs to be in Newmarket before 9:00 a.m. What time should she get the bus from Thetford?

3. How long does the 11a bus take from Thetford to London?

4. Which bus is the fastest from Norwich to Cambridge?

5. Arlo gets the bus from Thetford to visit his friend in Cambridge. It is delayed for 20 minutes due to traffic and he gets there at 09:54. What time did he get on the bus and how long did he spend on it?

6. Effie says, "The quickest bus from Thetford to London is the 11b."
Do you agree? Explain your answer.

7. Shamina says, "It took me 155 minutes to get from Thetford to London."
Which bus did she take?

8. What is the shortest journey you can make on the timetables?

Bus Timetable Answers

1. Daryl takes the bus and arrives at Cambridge at 9 o'clock. Which bus does he take?
bus 11a
2. Hanya needs to be in Newmarket before 9:00 a.m. What time should she get the bus from Thetford?
07:55
3. How long does the 11a bus take from Thetford to London?
departs Thetford 07:55
arrives London 10:32
time taken: 2 hours 37 minutes
4. Which bus is the fastest from Norwich to Cambridge?
Bus 11a takes 1 hour 50 minutes.
Bus 11b takes 1 hour 48 minutes.
Bus 11c takes 1 hour 37 minutes.
Bus 11c is the fastest bus from Norwich to Cambridge.
5. Arlo gets the bus from Thetford to visit his friend in Cambridge. It is delayed for 20 minutes due to traffic and he gets there at 09:54. What time did he get on the bus and how long did he spend on it?

The bus was delayed by 20 minutes and got there at 09:54, so it should have arrived at 09:34. This means he was on bus 11b.

He got on the bus at 08:30 and got off at 09:54, so he spent 1 hour and 24 minutes on the bus.

Bus Timetable Answers

6. Effie says, "The quickest bus from Thetford to London is the 11b."
Do you agree? Explain your answer.

Effie is wrong. Bus 11b takes 2 hours 35 minutes. Bus 11a is slower, at 2 hours 37 minutes, but bus 11c is faster, at only 2 hours 20 minutes.

7. Shamina says, "It took me 155 minutes to get from Thetford to London."
Which bus did she take?

155 minutes = 2 hours 35 minutes

Shamina took bus 11b.

8. What is the shortest journey you can make on the timetables?

The shortest journey you can make is from Newmarket to Cambridge on bus 11a, which takes 31 minutes.

Disclaimer: We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.